Collaborative Truck-UAV Delivery Routing Optimization Under Dynamic Weather Conditions and Customer Demands

Quan Yuan, Ruitai Li, Benkuan Zhou, Jing Lu, Menglan Hu, Pan Lai, Yingchao Zhao, and Xiao Zhang

***Abstract*— Unmanned Aerial Vehicles (UAVs) have emerged as core tools for logistics in smart cities, demonstrating significant potential due to their efficiency and flexibility. However, their capabilities in long-distance deliveries are constrained by payload and battery limitations. Additionally, most existing studies focus on route planning in static environments and largely overlook practical factors such as fluctuating winds and customer demands, all of which significantly impact UAV logistics efficiency. Thus, this paper explores a collaborative truck-UAV delivery model and develops an Adaptive Collaborative Logistics Algorithm (ACLA) for real-time path planning. To be specific, the model employs Normal-Weibull mixture and von Mises distribution to capture the stochastic nature of wind speed and direction, enabling real-time adjustments to UAV flight speeds for optimized route planning. Customer demands fluctuations are modeled by using Poisson distribution and associating a density clustering approach to dynamically redefine UAV service areas. To further enhance routing and scheduling efficiency, the proposed algorithm integrates a genetic algorithm with tabu search to ensure efficient task allocation and route planning under varying environmental conditions. The simulation results demonstrate that the proposed method performs exceptionally well in complex dynamic environments, significantly reducing total UAV energy consumption and delivery time while improving overall logistics efficiency.**

***Index Terms*— UAV path planning, winds, fluctuating customer demands, logistics, truck-UAV collaboration.**

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I. Introduction

With the booming of global e-commerce, the express logistics industry is confronted with increasing demands and challenges [1]. Traditional ground delivery is inefficient in speed, cost, and environmental impact. While UAVs offer a promising alternative by avoiding traffic and reducing last-mile costs [2], their range and payload are limited by battery constraints [3], and they struggle with dynamic demand alone [4]. Trucks, though suited for long hauls, face efficiency loss due to road networks [5]. Neither mode alone adequately meets diverse delivery needs, making truck–UAV collaboration an emerging research focus [6] that combines the strengths of both systems.

Current truck-UAV delivery research predominantly focuses on path planning efficiency [7], while overlooking dynamic environmental factors like wind effects [8]. For instance, studies optimize routes under dynamic demand but ignore wind impacts [9], or consider wind speed and payload while neglecting dynamic wind variations and time windows [10]. Although some works address wind in path planning, they assume static conditions and cannot handle real-time cross-winds [11]. Thus, models adaptable to dynamic winds and fluctuating customer demands remain needed.

Energy consumption critically constrains UAV performance in truck–UAV joint delivery [12], yet existing energy-optimization studies often overlook complex dynamic factors such as shifting wind speeds and fluctuating customer demand. For instance, while [13] examined wind effects on UAVs, they omitted dynamic wind conditions and service time windows; though [14] incorporated wind and energy constraints in path planning, they ignored demand variations and payload limitations. Thus, developing energy-optimization models that simultaneously address dynamic winds, variable demand, payload, and battery capacity remains essential to enhancing joint delivery efficiency.

This paper addresses truck–UAV collaborative routing, a complex two-echelon vehicle routing variant [15], under dynamic wind and fluctuating demand. We propose ACLA integrating a Hybrid Genetic Algorithm (HGA) and tabu

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search to minimize energy use and maximize efficiency. Key contributions are summarized as follows.

- To capture the stochastic nature of dynamic wind conditions and customer demands, the Normal-Weibull and von Mises distributions are used to characterize wind speed and direction variability, while the Poisson distribution models the randomness of customer requests. Based on these formulations, a collaborative multi-UAV-truck path planning model is established, aiming to minimize overall task completion time and UAV energy consumption through a global optimization strategy.
- To address the comprehensive optimization challenge of multi-UAV-truck collaborative logistics under dynamic conditions, ACLA is developed and validated. In the static phase, HGA generates the initial routing plan, while in the dynamic phase, tabu search is incorporated to accommodate environmental variations, thereby achieving joint optimization of routing and scheduling.
- Simulation results demonstrate that ACLA achieves outstanding performance in complex path planning scenarios. Compared with classical algorithms, such as greedy algorithms and KGA, ACLA shows superior adaptability and performance.
- Validated through both simulations and field experiments in the Guanggu Square District of Wuhan, the proposed model and ACLA exhibit strong adaptability to increasing customer demand, expanding service regions, and varying UAV fleet sizes.

This paper is structured as follows. Section II reviews UAV path planning and collaborative logistics. Section III formulates the truck-UAV delivery problem under dynamic winds and fluctuating demands. Section IV details the proposed algorithm framework, including pre-optimization and real-time processes. Section V presents simulation results and analysis, while Section VI concludes with key findings.

II. Related Work

Numerous research works address the civilian applications of UAVs. Banafaa et al. [16] elaborated on 5G technologies associated with UAVs. Cheng et al. [17] conducted a review of low-altitude UAVs within the Internet of Things (IoT) for data-related tasks. Hegde and Ghose [18] emphasized the utilization of UAVs in agriculture and transportation. Research indicates that UAVs, particularly in parcel delivery, possess significant potential.

A. Multi-UAVs Truck Collaborative Delivery Problems

Recent papers highlight truck-UAV cooperation to overcome UAV range limits [19], dynamic routing with Monte Carlo and LNS methods [20], and multi-UAV coordination via hybrid algorithms [21]. While these efforts improve delivery range and efficiency, they largely overlook dynamic environmental factors is a gap addressed by our model.

B. The Truck-UAVs Combined Routing Problems With Consideration of Time-Varying Road Travel Times

Existing papers on truck-UAV path optimization have considered time-varying road travel times but remain limited in key aspects. Wei et al. [22] incorporated time-varying road networks yet overlooked wind field effects on UAVs. Long et al. [23] proposed a dynamic coordination strategy for emergency scenarios but did not model UAV-demand coupling. Similarly, van Steenbergen et al. [24] used stochastic programming for truck travel uncertainty but ignored UAV scheduling. Collectively, these works prioritize truck-side optimization and lack a holistic framework accounting for UAV environmental adaptability.

C. Path Optimization Problem With Multiple Depots and Time Windows Constraints

Existing multi-depot path optimization studies exhibit notable limitations. Wang et al. [25] developed a cross-depot resource-sharing method that reduces cost but overlooks UAV collaboration. Li et al. [26] focused on multi-center reverse logistics, differing from our forward distribution model. While Wang et al. [27] addressed dynamic demand across depots, they ignored environmental factors like wind. Collectively, these works lack integrated optimization of trucks and UAVs under multiple dynamic constraints a gap this paper aims to fill.

D. Truck-UAV Cooperative Routing Problems Under Dynamic Wind

Research on truck–UAV routing under dynamic wind remains limited, despite its significant impact on UAV energy consumption. While Dorling et al. [28] established a linear energy relation for hovering UAVs, Sorbelli et al. [11] first incorporated wind into energy-aware trajectory planning. Momeni et al. [29] and Jung and Kim [30] further developed wind-aware path planning models. However, these studies predominantly assume constant wind conditions and lack real-time adaptability.

E. Truck-UAV Cooperative Routing Problems Under Dynamic Customer Demands

Dynamic customer demands render UAV delivery path planning more intricate. Existing research provides certain solutions. Pan et al. [31] employed a digital twin driven approach for industrial park orders. Zhang et al. [32] analyzed collaborative delivery under dynamic demands and environments. Dayarian and Savelsbergh [33] proposed a delivery model where trucks handle preorders and UAVs address new demands, yet it fails to fully exploit UAV flexibility. Ulmer and Thomas concentrated on same day delivery UAV path planning without optimizing time and energy [34]. In summary, current research overlooks the combined impact of demand and wind variations, thus having difficulty meeting real-world logistics requirements.

F. Research Gaps

Building on existing studies, this paper highlights three gaps in UAV delivery research including energy models that overlook wind dynamics, insufficient analysis of demand-environment interactions, and limited work on

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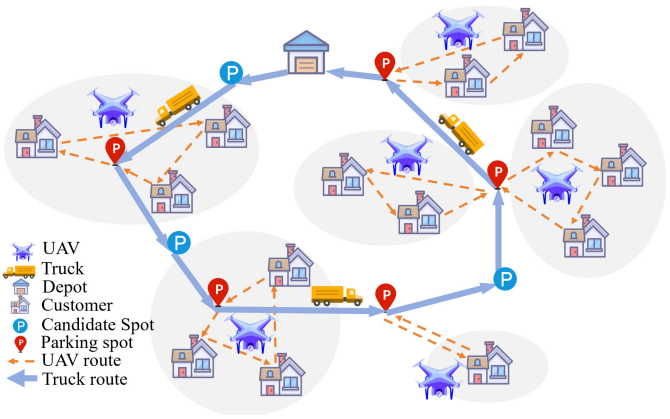


Fig. 1. Schematic diagram of the multi-UAVs-truck cooperative parcel delivery problem.

multi-UAV coordination in dynamic contexts. To address these, we develop a two-stage model incorporating routing and dynamic demand under wind uncertainty, with a hybrid planning strategy and truck-UAV coordination to enhance operational efficiency.

III. Problem Formulation

This research focuses on a mission scenario featuring a collaborative package delivery system using UAVs and trucks. UAVs have limited endurance, constraining long-distance deliveries, so trucks support parcel delivery to offset this drawback. The system is modeled as a one-way road transportation network $G = (V, E)$, where $E = \{(i, j) \mid i, j \in V\}$ represents the set of edges, $V = \{0, 1, \ldots, n\}$ denotes the set of nodes, and $V_1 = V/\{0\}$ represents the set of customer points, with 0 denoting the warehouse. All packages start at the warehouse. Trucks, carrying UAVs and customer packages, follow predetermined routes, stopping at specific sites. At each stop, multiple UAVs are launched from the truck to serve nearby customer clusters. After meeting the delivery needs of all customers in a cluster, the UAVs return to the truck, which then returns to the warehouse. The k-th customer cluster at time t is denoted as $C_k(t)$, consisting of several customers $i \in C_k(t)$. Specifically, $C_k(t) = \{i_1, i_2, \ldots, i_{n_k}\}$, where n_k is the number of customers in $C_k(t)$, and each customer is served by one UAV.

During UAV operations, wind speed critically influences flight efficiency. While favorable wind directions enhance airspeed, unfavorable ones reduce it, making dynamic wind conditions a key factor in energy consumption. Parcel weight further constrains UAV energy use. However, prioritizing energy savings may compromise delivery speed, which is critical to customer satisfaction. This paper therefore accounts for battery and payload limits, and optimizes the coordination of multiple UAVs and a truck to minimize total delivery time.

In summary, this paper utilizes mathematical methods to tackle a bi-objective optimization problem. The weighted average of minimizing the energy consumption of UAVs and the total delivery time to meet all customer demands is derived as a single-objective function, as illustrated in Fig. 1.

TABLE I
Notation and Terminology

Notation	Definition
W_1	weight of the UAV's frame structure
W_2	Weight of the UAV's battery
D_{max}	the maximum payload of the UAV
L_{max}	the maximum hovering distance of the UAV
q	payload weight carried by the UAV
g	gravitational acceleration
ρ	air density
ς	area of the rotor disk
n_0	number of rotors on the UAV
v_d	initial velocity of the UAV
θ_d	initial flight direction of the UAV
v_f	wind speed
θ_f	wind direction
v_t	final velocity of the UAV under wind influence
θ_t	final flight direction of the UAV under wind influence
$D(t)$	set of UAVs
$C(t)$	set of customers
$K(t)$	number of clusters
t	global timeline variable
T_{ij}^d	UAV flight time from i to j
q_{ij}^d	remaining payload from i to j
E_{total}	total energy consumption of the UAV system
T_{total}	total delivery time of the UAV system
P_k	customer sequence in cluster k
P_s^d	UAV d's customers at stop s
C_s	customers served at stop s
S	the set of truck stops
D_s	UAVs launched at stop s

Table I explains the symbols and terms used in this paper. We propose an approach to jointly optimize UAV flight paths, truck routes, and parcel delivery task allocation under dynamic customer demand scenarios. The problem is divided into the following aspects.

A. UAV Energy Consumption

The UAV energy consumption model follows [28]:

$$p = (W_1 + W_2 + q)^{\frac{3}{2}} \sqrt{\frac{g^3}{2\rho\varsigma n_0}} \tag{1}$$

Since some parameters are constant, the power consumption can be simplified as:

$$p = \alpha (Q + q)^{\frac{3}{2}} \tag{2}$$

where $\alpha = \sqrt{\frac{g^3}{2\rho\varsigma n}}$, $Q = W_1 + W_2$ are constants. Therefore, UAV power consumption depends solely on the cargo weight. The energy consumption e for flying a distance l is:

$$e = pt = \alpha (Q + q)^{\frac{3}{2}} \frac{l}{v_t} \tag{3}$$

When minimizing the energy consumption of UAVs, the weight of the UAV load and the flight speed should be taken

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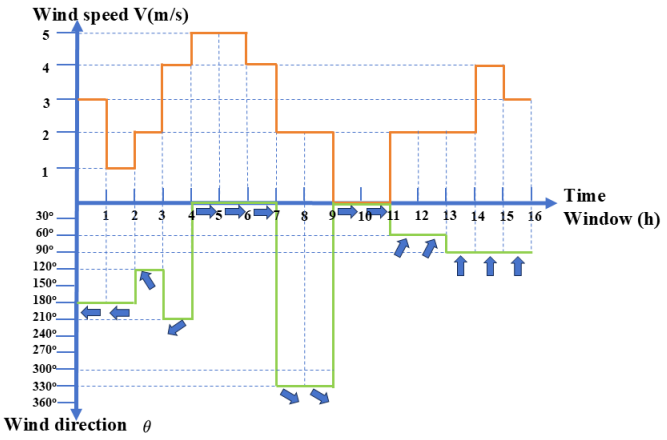


Fig. 2. Examples of changes in wind speed and direction.

into account. In this research, parametric quantification is employed during the flight process. The direction of the wind is modeled using the von Mises mixture distributions and the speed of the wind is modeled with the Normal-Weibull distributions, which are prevalent in meteorology [38]. The operational protocols are updated hourly in accordance with the standards of the National Weather Service (NWS) [39], striking a balance between real-time feasibility and practical monitoring requirements. Examples of changes in wind speed and direction are shown in Fig. 2.

Since coordinated delivery of UAVs and trucks involves multiple take-offs and landings, and UAV distribution among multiple clusters, this article uses local dynamic wind speed to more accurately simulate the change of wind speed within a short period of time. The dynamic change of wind speed v_f is satisfied with the change of wind speed over time in different customers i and j:

$$f\left(v_f, t\right) = \omega\left(t\right) \cdot f_N\left(v_f, \mu\left(t\right), \sigma\left(t\right)\right) + \left(1 - \omega\left(t\right)\right) \cdot f_W\left(v_f, k\left(t\right), c\left(t\right)\right) \tag{4}$$

$$f_N\left(v_f, \mu\left(t\right), \sigma\left(t\right)\right) = \frac{1}{\sigma\left(t\right)\sqrt{2\Pi}} e^{-\frac{\left(v_f - \mu(t)\right)^2}{2\sigma(t)^2}} \tag{5}$$

$$f_W\left(v_f, k\left(t\right), \lambda_t\right) = \frac{k\left(t\right)}{\lambda\left(t\right)}\left(\frac{v_f}{\lambda\left(t\right)}\right)^{k(t)-1} e^{-\left(\frac{v_f}{\lambda(t)}\right)^{k(t)}} \tag{6}$$

Wind direction θ_f Satisfaction:

$$f\left(\theta_{ij}\left(t\right)|\mu, k\right) = \frac{e^{k\cdot\cos\left(\theta_{ij}(t)-\mu\right)}}{2\pi I_0\left(k\right)} \tag{7}$$

$$\theta_{ij}\left(t\right) = \theta_f \tag{8}$$

Since the speed of the UAV will fluctuate due to the influence of wind, the vector relationship of the speed of the UAV is obtained as shown in Fig. 3.

The speed relationship of a UAV under the influence of wind satisfies:

$$\begin{bmatrix} v_t\cos\theta_t \\ v_t\sin\theta_t \end{bmatrix} = \begin{bmatrix} v_d\cos\theta_d \\ v_d\sin\theta_d \end{bmatrix} + \begin{bmatrix} v_f\cos\theta_f \\ v_f\sin\theta_f \end{bmatrix} \tag{9}$$

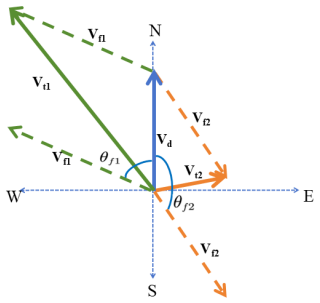


Fig. 3. UAV speed variation map.

When UAV departs from the customer$i$ to deliver a package to the customer j, it is affected by the wind speed $w_{ij}\left(t\right)$:

$$T_{ij}^d = \int_0^{L_{ij}} \frac{1}{v_t} dL \tag{10}$$

When evaluating the influence of the UAV load on energy consumption, we take into account its dynamic variation at every customer location. As the UAV delivers packages for all customers within a group starting from the truck stop, its load progressively decreases after serving each customer in that group.

$$q_{ij}^d = \sum_{h\in P_k, h\geqslant j} q_h \tag{11}$$

where $x_{ij}^{d,k}$ are decision variables. When $x_{ij}^{d,k} = 1$ indicates that the UAV d changes from the customer i to the customer j within the group k, otherwise $x_{ij}^{d,k} = 0$. The total energy consumption of a UAV is the aggregate of the energy utilized in each path segment.

$$E_{total} = \sum_{k\in K(t)} \sum_{(i,j)\in P_k} \alpha\left(Q + q_{ij}^d\right) T_{ij}^d \cdot x_{ij}^{d,k} \tag{12}$$

B. Generation of Dynamic Customer Demand

During the distribution process, customers within a cluster have the potential to generate dynamic demands at any moment. Within a specific time frame, new demands might arise, and existing demands could be canceled. The schematic diagrams are presented in Fig. 4(d) and Fig. 4(e).

The set of customers is defined as $C\left(t\right) = \left\{C_1\left(t\right), C_2\left(t\right), \ldots, C_{n(t)}\left(t\right)\right\}$, where $C\left(t\right)$ represents the set of customers at time t, satisfying the following relationship:

$$C\left(t\right) = \bigcup_{k=1}^{K(t)} C_k\left(t\right) \tag{13}$$

At time t, $n\left(t\right)$ represents the number of customers, which undergoes dynamic changes. The number of new customers, $N_{new}\left(t\right)$ and the number of customers terminating the service $N_{out}\left(t\right)$ adhere to a Poisson distribution with parameters λ_{new} and λ_{out} respectively. The precise locations of newly added customers can be either randomly distributed or dictated by

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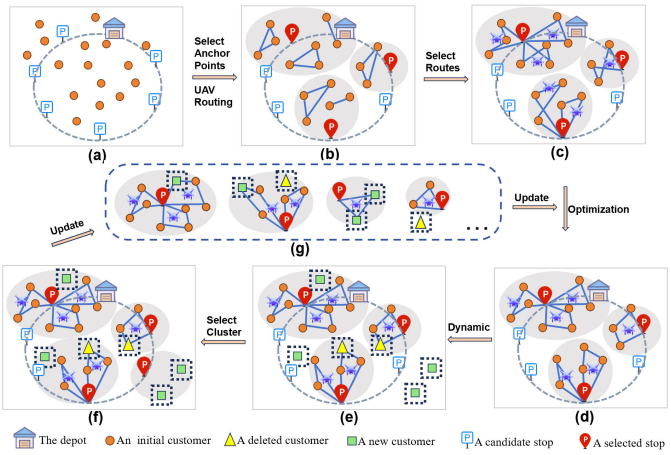


Fig. 4. ACLA process for handling dynamic customer demands.

specific rules, like a uniform distribution across the map or proximity to existing clusters.

$$P\left(N_{new}\left(t\right)=k_0\right)=\frac{\lambda_{new}^{k_0}e^{-\lambda_{new}}}{k_0!} \tag{14}$$

If the location of the added customer $c_{new}\left(t\right)$ is close to $C_k\left(t\right)$, it is added to the cluster:

$$C_k\left(t\right)=C_k\left(t\right)\bigcup\left\{c_{new}\left(t\right)\right\} \tag{15}$$

If it is too far from all clusters, add new clusters:

$$K\left(t+1\right)=K\left(t\right)\bigcup\left\{C_{K(t)+1}\left(t\right)\right\} \tag{16}$$

At the moment t some customers may cancel the demand:

$$P\left(N_{out}\left(t\right)=k_0\right)=\frac{\lambda_{out}^{k_0}e^{-\lambda_{out}}}{k_0!} \tag{17}$$

The rules for selecting clients for removal are based on uniform randomization:

$$C_k\left(t\right)=C_k\left(t\right)-\left\{c_{removed}\left(t\right)\right\} \tag{18}$$

Remove a cluster $C_k\left(t\right)$ if the number of clients in that cluster drops to 0:

$$K\left(t+1\right)=K\left(t\right)-\left\{C_k\left(t\right)|n_k\left(t\right)=0\right\} \tag{19}$$

Dynamic changes in the set and total number of clients are satisfied:

$$C\left(t+1\right)=C\left(t\right)\bigcup C_{new}\left(t\right)-C_{removed}\left(t\right) \tag{20}$$

$$n\left(t+1\right)=n\left(t\right)+N_{new}\left(t\right)-N_{out}\left(t\right) \tag{21}$$

C. Total Delivery Time for Customer Requirements

At each truck stop, multiple UAVs take off from the truck to serve different clusters. We define the service time of a truck stop as the time taken by the UAV with the longest service duration. Consequently, the total delivery time is the sum of the longest service times at each truck stop. To ensure that all customers are served at each stop, the following relationship must be satisfied:

$$\bigcup_{s\in S}C_s=C \tag{22}$$

Get the total time for completion of all customer requirements of the system:

$$T_{total}=\sum_{s\in S}\max_{d\in D_s}\left(\sum_{(i,j)\in P_s^d}t_{ij}^d\right) \tag{23}$$

Regarding the total driving time of the truck, assume that the truck is driving at a constant speed v_{truck}. Denote the decision variable as p_{kz}, where $p_{kz}=1$ indicates that the truck drives from parking point k to parking point z, otherwise, $p_{kz}=0$. Let s_{kz} represent the distance from parking point k to parking point z. Then the total driving time of the truck can be expressed as:

$$T_{truck}=\frac{\sum_{k,z\in S}s_{kz}\cdot p_{kz}}{v_{truck}} \tag{24}$$

The total global time is:

$$T_{global}=T_{total}+T_{truck} \tag{25}$$

D. Optimization Objectives

In this paper, the final bi-objective optimization formula is as follows by considering the flight energy consumption of the UAV and total delivery time of the system.

$$\min\left(E_{total},T_{global}\right) \tag{26}$$

Weighted method converted it to a single target.

$$\min Z=\lambda_1E_{total}+\lambda_2T_{global} \tag{27}$$

where $0\leqslant\lambda_1\leqslant1$ and $0\leqslant\lambda_2\leqslant1$ are weighting factors that satisfy the condition $\lambda_1+\lambda_2=1$.

$$\text{s.t.}\quad\bigcup_{s\in S}\bigcup_{d\in D_s}P_s^d=C \tag{28}$$

$$\sum_{j\in C}x_{0j}^{d,k}=\sum_{j\in C}x_{i0}^{d,k}=y_k^d \tag{29}$$

$$\sum_{j\in P_k}q_jx_{ij}^{d,k}\leqslant D_{max},\quad\forall i\in C,d\in D \tag{30}$$

$$\sum_{i\in C\bigcup\{0\}}\sum_{j\in C\bigcup\{0\}}L_{ij}\cdot x_{ij}^{d,k}\leqslant L_{\max},\ \forall d\in D,\quad\forall k\in K \tag{31}$$

Constraint (28) guarantees that each customer is served exactly once by one UAV. Constraint (29) ensure that each UAV departs from and returns to the same truck parking location. where y_k^d is the decision variables, and $y_k^d=1$ indicates that UAV d takes off from the truck's k-th stopping point, otherwise, $y_k^d=0$. Constraint (30) guarantees that the UAV's payload on any path must not exceed its maximum capacity. Constraint (31) guarantees that the length of a route cannot exceed the limit of hovering distance of a UAV.

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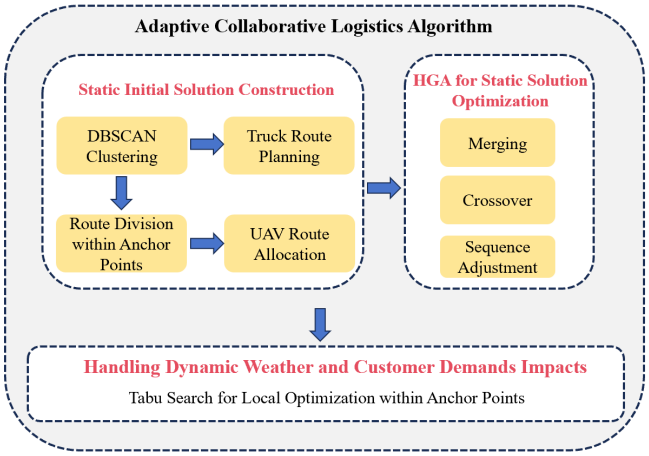


Fig. 5. Flowchart of the ACLA.

IV. Algorithm Design

In the truck-UAV collaborative system, trucks only transport UAVs to anchor points without performing parcel delivery. Although minor variations in truck speed slightly influence travel time and overall cost variability, they do not significantly alter anchor positioning or route planning. In contrast, weather-induced UAV speed changes can disrupt or terminate planned paths, requiring substantial real-time adjustments to routes and task assignments.

As illustrated in Fig. 5, the ACLA framework optimizes truck-UAV coordination through a two-phase strategy. The static phase employs a Hybrid Genetic Algorithm (HGA) to determine anchor points, plan UAV routes, and schedule truck paths under stable conditions, initialized by a greedy heuristic. The dynamic phase adapts to real-time wind and demand variations using periodic rescheduling and tabu Search, enabling robust task execution in dynamic environments.

A. Collaborative Delivery Algorithm Under Static Conditions

Under static conditions, we introduce HGA for logistics planning. While the genetic algorithm offers global search capability, it suffers from slow convergence and a tendency to stagnate locally. The HGA overcomes these limitations by hybridizing it with local search, significantly accelerating convergence and improving solution quality. The static-phase HGA follows the procedure below.

1) Initial Solution Construction:

a) Anchor Point Selection Based on DBSCAN Algorithm: This paper addresses customer clustering for parking anchor point allocation with geographically dispersed, non-spherically distributed customers under dynamic demand. Unlike K-means—which requires preset cluster numbers and assumes isotropic data [40]—DBSCAN better suits real-time needs by forming density-based arbitrary-shaped clusters, automatically delineating service zones while suppressing noise. The proposed method thus employs DBSCAN for anchor point selection in the initial static task set $P_0 = \{p_1, p_2, \ldots, p_n\}$ in region Ω. The objective is to allocate one anchor point for each task point in Fig. 4(b). A candidate anchor point set A is evenly distributed in the map according to preset rules. The coverage condition is defined as follows: if the distance between a task point p_i and an anchor point a_i is less than or equal to half of the maximum UAV flight distance $D_{\max}$ and within the maximum payload range $L_{\max}$, then the anchor point is considered to cover the task point. The set of task points covered by a_i is $C_{P_0}^{a_i}$. The performance of an anchor point is evaluated by the task point coverage rate $N_{\alpha}^{a_i}$, which is the number of covered task points, and is calculated as:

$$N_{\alpha}^{a_i} = \frac{|C_{p_0}^{a_i}|}{|p_0|} \tag{32}$$

DBSCAN partitions customers for UAV truck collaborative services through density-based clustering. Initially, all customer points are labeled as unvisited. Commencing from an unvisited point, it examines the neighboring points within the radius Eps. If the number of neighbors is greater than or equal to $minPts$, the point is identified as a core point, and its neighbors are incorporated into the current cluster. Subsequently, the cluster expands by scrutinizing the neighbors of the points within the cluster until no additional points can be added. This process groups all customers into clusters, with each cluster serving as a UAV flight task area. For each cluster, the UAV takeoff and recovery points are ascertained to optimize the collaboration between the truck and the UAV and to construct the flight segments. The pseudocode of the algorithm is presented in Algorithm 1.

Algorithm 1 DBSCAN Clustering Algorithm

Require: $P_0 = \{p_1, p_2, \ldots, p_n\}$ (Set of all customers), anchor set $A = \{A_{\text{start}}, a_1, a_2, \ldots, a_t\}$, Eps (Neighborhood radius), $minPts$ (Minimum points per cluster)
Ensure: T (Clusters containing all customers)

```
Initialize all points in P as unvisited
T ← ∅
while there exists an unvisited point p_i ∈ P do
    Mark p as visited
    Find N_Eps(p_i) = {q ∈ P | d(p_i, q) ≤ Eps} (Neighborhood of p_i)
    if |N_Eps(p_i)| ≥ minPts then
        Create a new cluster C_new ← {p_i}
        for all q ∈ N_Eps(p_i) do
            if q is unvisited then
                Mark q as visited
                Add N_Eps(q) to N_Eps(p_i)
            end if
            Add q to C_new
        end for
        C ← C ∪ {C_new}
    end if
end while
for C_j ∈ C do
    Find anchor a_q ∈ A that minimizes the total distance to all points in C_j
    T ← T ∪ {(a_q, C_j)}
end for
return T
```

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b) UAV Route Planning for Each Cluster: After parking anchor points are divided, each a_i has a task set T_i. As illustrated in Fig. 4(b), we formulate the UAV routes for each T_i taking into account the endurance and payload limitations. First, we initialize an empty set of routes, a single route, and the starting position. At each step, a task point from T_i is randomly selected and appended to the route. Subsequently, we verify whether the flight from the anchor point, through the route, and back to the anchor point surpasses the UAV's maximum flight distance D_{max}, or whether the total package weight within the route exceeds the maximum single-load L_{max}. In the event that either of these limitations is reached, the task point is removed from the route, and the route is incorporated into the set of routes. By this means, the UAV is able to serve as many task points as possible in a single flight while adhering to the constraints. After several iterations, once T_i is exhausted, the set of UAV flight routes for that particular anchor point is outputted.

c) UAV Flight Route Allocation within Customer Clusters: After partitioning flight routes at a docking anchor point in Fig. 4(c)), it is essential to efficiently allocate these routes to UAVs to achieve maximum task efficiency. The objective is to minimize the anchor-point task time while all UAVs operate simultaneously. Initially, each route length l_{ri} is compared with the remaining flight ranges e_{ui} of all UAVs. If $l_{ri} \leq e_{ui}$ for a UAV u_i, route r_i is assigned to u_i, added to its task set, and removed from the unassigned list. Upon mission completion, all UAVs are marked as completed with their positions and ranges reset. Unassigned routes are then sequenced by total flight distance and sequentially allocated to the UAV with the currently shortest assigned route length, ensuring full task distribution.

d) Truck Route Optimization: Truck routing optimization aims to minimize travel time, yet conventional algorithms often converge to local optima and demand sensitive parameter tuning. [41]. This paper adopts a two-step iterative heuristic where the Lin-Kernighan heuristic (LKH) first identifies an optimal truck route, then refines it via random perturbation and 2-OPT strategies. Over n iterations, any route improving the objective value updates the incumbent solution. The LKH method accelerates convergence by dynamically refining local paths using an enhanced LKH.

2) Iterative Optimization of Static Solutions: Building on the greedy solution S developed for anchor selection, task division, and route planning, this paper introduces HGA to further refine the results. By combining genetic search with local optimization, the HGA achieves effective global exploration and local refinement, proving highly capable in handling complex routing problems. As shown in Fig. 6, the algorithm iterates through population initialization, genetic operation, offspring optimization, and population update.

a) Initial Population Generation: The HGA improves upon standard genetic algorithms by using heuristic methods to generate high-quality initial populations, thereby accelerating convergence. A maximum population size M_{max} is defined, and feasible solutions derived from the previously described greedy strategy are incorporated into the global set while monitoring its size.

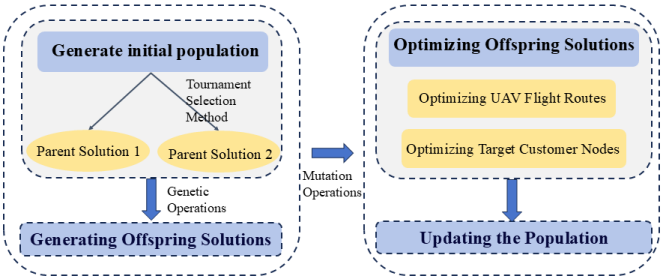


Fig. 6. The schematic diagram of the HGA.

b) Genetic Operations: Genetic algorithm is centered around common operations such as selection, crossover, and mutation. In this paper, genetic operations combine favorable gene segments from parent solutions S_1 and S_2 to generate offspring, as illustrated in Algorithm 2. Roulette wheel selection is employed as the selection operator, which requires no additional parameters. It assigns selection probabilities proportional to individual fitness values, favoring fitter individuals while preserving population diversity through its stochastic nature, thereby enhancing algorithmic robustness.

Algorithm 2 Crossover Algorithm

Require: Solution P_1, Solution P_2
Ensure: Crossed solution C

$R_{P1} \leftarrow$ Extract all routes from P_1
$R_{P2} \leftarrow$ Extract all routes from P_2
$R_{Total} \leftarrow R_{P1} \cup R_{P2}$
$A_{P1} \leftarrow$ Extract all anchor points from P_1
$A_{P2} \leftarrow$ Extract all anchor points from P_2
$A_{Total} \leftarrow A_{P1} \cup A_{P2}$
$R_{final}, A_{final} \leftarrow \emptyset$
while $R_{Total} \neq \emptyset$ **do**
    Select the route r_i with the lowest cost from R_{Total}
    $R_{final} \leftarrow R_{final} \cup r_i$
    $R_{Total} \leftarrow R_{Total} \setminus r_i$
    Remove all route segments containing r_i from R_{Total}
end while
while $R_{final} \neq \emptyset$ **do**
    Select an anchor point a_i from A_{Total} that covers most customers
    $A_{final} \leftarrow A_{final} \cup a_i$
    $R_{temp} \leftarrow$ Select all route sets R_{ai} in R_{final} covered by a_i
    for $r_i \in R_{temp}$ **do**
        $R_{final} \leftarrow R_{final} \setminus r_i$
    end for
end while
return Crossed solution C

c) Optimizing Offspring Solutions: A local search algorithm enhances offspring solutions by jointly optimizing UAV routes and customer sequences during the HGA mutation phase, primarily through merging short routes, splitting long ones, and reordering customer visits.

- Merging Short Routes: Using route and anchor point sets as inputs, the algorithm iteratively selects two routes

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shorter than a tunable distance via tournament selection. Routes are merged provided the combined route satisfies distance constraints and yields a net length reduction. This process repeats until the maximum iteration count is reached, outputting the optimized routes. For a route r_i, its neighboring route $Neigh(r)$ is defined as:

$$Neigh\left(r_i\right)=\left\{r_x \left| \max\left(a_{n_i n_x}\right)<d, n_i \in r_i, n_x \in r_x\right.\right\} \tag{33}$$

- Long Route Splitting: Input the route set and anchor points. Employ the roulette wheel selection method to select a long route. Split this long route at the two most distant adjacent task points to obtain two shorter routes. Continuously repeat this process until the maximum iteration number is reached. Subsequently, output the optimized route set.
- Adjusting Customer Visit Order in Routes: Given a set of routes, the algorithm randomly selects a UAV flight route $route$. It chooses a task point from $route$. If incorporating this point into the final route $route_{final}$ results in the smallest increase in the total length, the point is removed from $route$ and added to $route_{final}$. This operation is repeated until $route$ becomes empty. If the total length of $route_{final}$ satisfies the distance constraints and is shorter than that of $route$, $route_{final}$ substitutes $route$; otherwise, $route$ remains unchanged. The procedure continues until the maximum number of iterations is reached, thus generating an optimized set of routes.

d) Population Update: Finally, the population is updated by utilizing optimized offspring solutions. After several iterations, if the population size M surpasses the limit M_{max}, global solutions are sorted according to cost. The $|M|-M_{max}$ least effective solutions are eliminated to prevent overflow. When the maximum iteration number is reached, the optimal solution is selected.

3) Convergence Analysis of HGA: The HGA synergistically integrates global and local search, with the former maintaining population diversity and the latter accelerating refinement. As shown in Fig. 7, under identical experimental settings, the HGA demonstrates marked convergence superiority, stabilizing after around 120 iterations and achieving approximately 14% higher fitness than the standard Genetic Algorithm, which is prone to local optima and slow convergence. This collaboration enhances both search efficiency and solution quality.

B. Collaborative Delivery Algorithm for Trucks and UAVs Under Dynamic Conditions

1) Evaluation of Dynamic Change Indicators of Customer Demands: Accurately evaluating shifting customer demand is essential for dynamic routing optimization. A key metric for quantifying such demand dynamics is the proportion of new customers relative to the total customer base. Let $N_{new}(t)$ denote the number of new customers added at time t, which follows a Poisson distribution with parameter λ_{new}, and its probability mass function is $P(N_{new}(t)=k)=\frac{\lambda_{new}^k e^{-\lambda_{new}}}{k!}$, $k=0,1,2,\cdots$, and the value of $N_{new}(t)$ is determined through actual statistics. Let $N(t)$ be the total number of customers at time t, and its calculation is based on the initial number of customers $N(0)$ and is dynamically updated according to the increase of new customers and the cancellation of original customers, that is, $N(t)=N(0)+N_{new}(t)-N_{out}(t)$, where $N_{out}(t)$ is the number of customers canceling the service at time t and follows a Poisson distribution with parameter λ_{out}. Then the calculation formula of the ratio $R(t)$ of the number of new customers to the total number of customers is $R(t)=\frac{N_{new}(t)}{N(t)}$.

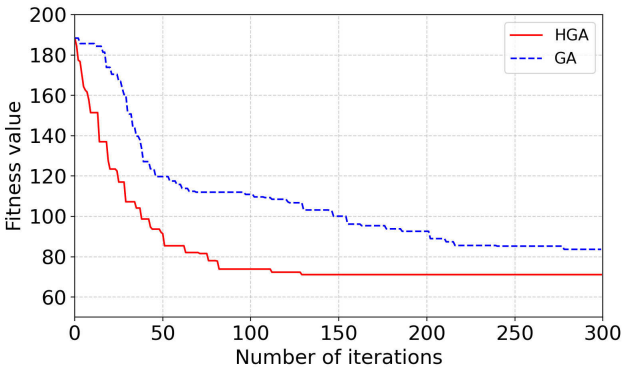


Fig. 7. The convergence of various algorithms in the static conditions.

2) Real-Time Scheduling Algorithm and Dynamic Level Assessment: To handle dynamic weather and customer demands—such as new orders, cancellations, or address changes—a real-time scheduling approach is introduced. The dynamic customer ratio, which grows over time, quantifies demand variability. Route adjustments are performed periodically within short cycle ξ. Starting from T_a with a statically planned route, vehicles initially follow this baseline. From T_b, dynamic time windows are segmented into intervals; each integrates updated demands and performs local re-optimization using current customer data.

3) Local Optimization Phase: During local optimization, the algorithm refines solutions at each anchor point by adjusting UAV routes and improving scheduling to minimize total task time. First, it identifies the nearest anchor to a task and adds the task point to the current-time customer set C at anchor point d_i. For the alterations in the customer set served by the UAV, Algorithm 3 is employed to reoptimize the UAV flight routes and the customer route allocation at the anchor. A tabu search algorithm modifies the UAV routes within the anchor and generates a locally optimized solution.

4) Global Optimization Phase: In the global optimization phase, unscheduled task points are first clustered via DBSCAN to group spatially proximate nodes, reducing inter-task travel distance and time. The instantaneous static problem for the current period is then resolved to minimize the total time to completion.

V. Performance Evaluation

To fully evaluate the effectiveness and robustness of the ALCA, this paper designs four experiments: multi-weather comparative tests, comparisons with competitors, sensitivity tests of the algorithm, and verifications in real scenarios.

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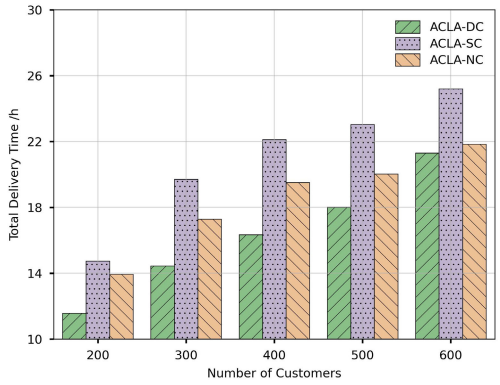
Algorithm 3 Tabu Search Optimization

Require: Initial Solution S with a single anchor point
Ensure: Optimized Solution S'

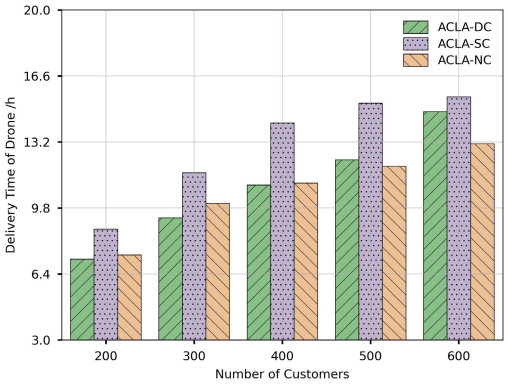
$S' \leftarrow S$
$\tilde{S} \leftarrow S$
$Tabu_list \leftarrow \emptyset$
while $max_iter > 0$ **do**
  $neigh \leftarrow \emptyset$
  $count \leftarrow 0$
  while $count < k1 + k2$ **do**
    if $count < k1$ **then**
      Randomly select two routes and randomly exchange a customer point from each route
    else
      Randomly select two routes, randomly select a customer point from one route and add it to the other route
    end if
    Re-sort the newly formed two routes
    Delete empty routes formed by the insertion operation
    Reassign routes to UAVs to generate the solution within a single anchor point
  end while
  $\tilde{S} \leftarrow$ Best solution in neighborhood
  Update the Tabu list based on $\tilde{S}$
  Calculate costs V' and $\tilde{V}$ of optimized solution S' and solution $\tilde{S}$
  if $\tilde{V} < V'$ **then**
    $S' \leftarrow \tilde{S}$
  end if
end while
return S'

Evaluation Metrics: Three metrics are adopted for evaluation: total time cost, total operational cost, and UAV operating time cost. Total time cost measures the complete duration from depot departure to return of both truck and UAV. In dynamic settings, routes are replanned following each change, necessitating repeated path planning. Simulations track vehicle states and positions to compute actual operating time. The unit cost of the truck is approximately 0.78$ per kilometer, and that of the UAV is around 0.078$ per kilometer [36]. Smaller values of the metrics signify lower costs, which demonstrates the higher efficiency of the algorithm.

Simulations Environment: In the simulation, customers are randomly positioned in a 50 × 50 km area, within which the coordinates of warehouses are randomly generated. These warehouses store parcels with different weights. We utilize three ARK-150 hexacopter UAVs for delivery [37] (the parameters are presented in Table II). The truck speed is set to fluctuate near 15 m/s under default parameters, simulating real-world conditions. We investigate how task point count, target area size, and UAV fleet size affect algorithm performance, with each parameter varied across a predefined range to evaluate its impact.



(a) Total Delivery Time



(b) Delivery Time of UAV

Fig. 8. The impact of different weather condition.

TABLE II
THE PARAMETERS OF UAV ARK-150

Parameters	Value
Empty weight	100 kg
Maximum payload	30 kg
Maximum charge	16 kWh
TAS	20 m/s

Benchmark Algorithms: Given the absence of directly comparable algorithms in the literature, we adapt two established methodsof the greedy algorithm and KGA as benchmarks. The greedy algorithm serves customers through multi-trip routing with rationally selected anchor points, while the KGA [35] employs K-means to determine launch locations and a GA to solve the truck TSP for minimal delivery time. Compared to these, our ACLA demonstrates clear efficiency gains under dynamic weather conditions, validating the advantage of its dynamic optimization scheme.

A. Comparison of Comparison of ACLA Under Different Weather Scenarios

The ACLA was tested under three weather scenarios including no change (NC), static conditions (SC), and dynamic conditions (DC). With customer counts varying from 200 to

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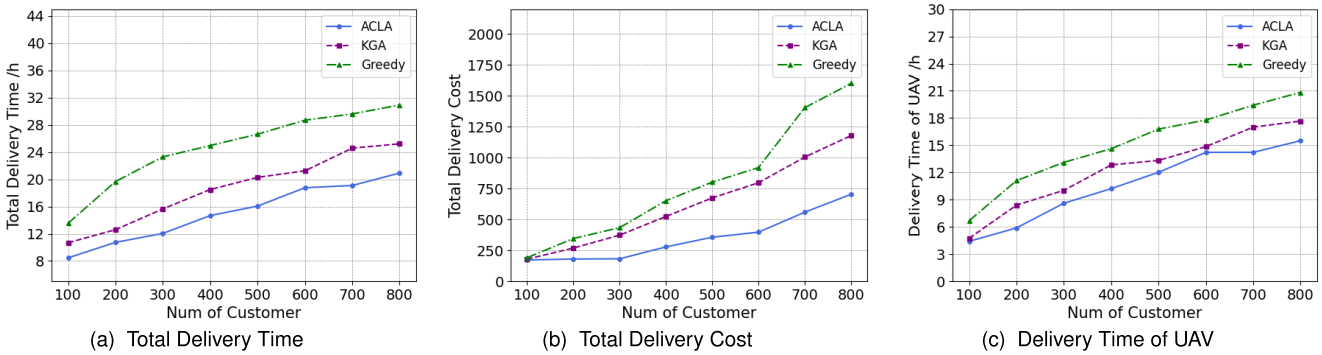


Fig. 9. The impact of different customer numbers.

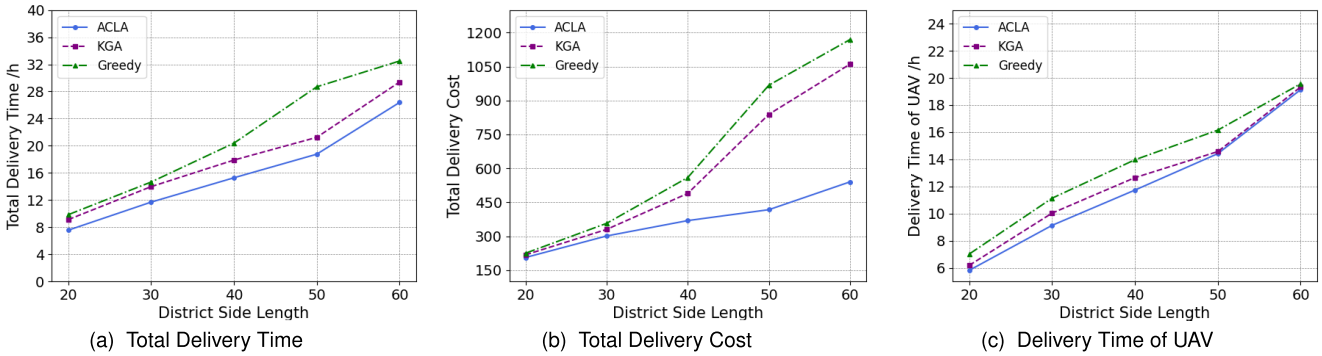


Fig. 10. The impact of different service area size.

600 and other parameters fixed, Fig. 8 shows the resulting delivery durations. Truck travel time was excluded as it depends primarily on anchor points rather than weather. Results indicate that delivery is fastest under NC due to stable UAV speeds, while in DC, UAVs dynamically adjust routes to leverage favorable winds, conserving energy and achieving optimal efficiency. This demonstrates that ignoring dynamic weather in route planning may compromise both delivery timeliness and customer satisfaction.

B. Comparison of ACLA With Competitors

1) The Impact of Different Customer Numbers: In Fig. 9 examines how task point volume (100-800) affects algorithm performance. In Fig. 9(a), all algorithms show linearly increasing time costs, steepest for the greedy method. The ACLA outperforms both benchmarks by dynamically adapting routes to wind and demand, validating its optimization efficacy. In Fig. 9(b), while total costs rise with task points, the ACLA's anchor point selection via DBSCAN mitigates truck travel growth, curbing cost increases. In Fig. 9(c), UAV flight time rises accordingly, yet the ACLA moderates this growth through energy-aware route adjustments that account for endurance and dynamic factors.

2) The Impact of Different Service Area Size : In Fig. 10 evaluates algorithm performance with 600 task points across varying target area sizes (20-60 km edge length). Time cost rises linearly with area size in Fig. 10(a), where ACLA outperforms both KGA and greedy algorithm by reducing vehicle travel distance, demonstrating particular efficiency advantage in larger regions.

Fig. 10(b) reveals growing cost differences with expanded work areas, where ACLA's parking point selection yields the lowest cost, demonstrating its adaptability across varying task areas. In Fig. 10(c), UAV flight time increases with target area size. While the greedy algorithm and KGA often settle for local optima, resulting in inefficient global paths and longer flights, ACLA groups proximate tasks into coherent routes, minimizing detours and reducing flight time.

3) The Impact of Different UAV Numbers: Fig. 11 examines UAV fleet size impact with 600 task points in a 50 × 50 km area. In Fig. 11(a), time cost initially decreases then plateaus as UAVs increase, indicating diminishing marginal returns. ACLA maintains the lowest time cost throughout. In Fig. 11(b), while added UAVs reduce delivery costs through parallel processing, excessive units diminish returns due to unbalanced task allocation. In Fig. 11(c), ACLA achieves the shortest flight times through efficient task clustering and path planning, outperforming the greedy algorithm's suboptimal route allocation.

C. Sensitivity Analysis of the ACLA

1) The Impact of Different Dynamic Changes and Adjustment Cycles : This investigation examines the performance of the algorithm under dynamic wind conditions and alterations in customer demand by changing the adjustment intervals from 0.25 to 3 hours, eeping other parameters

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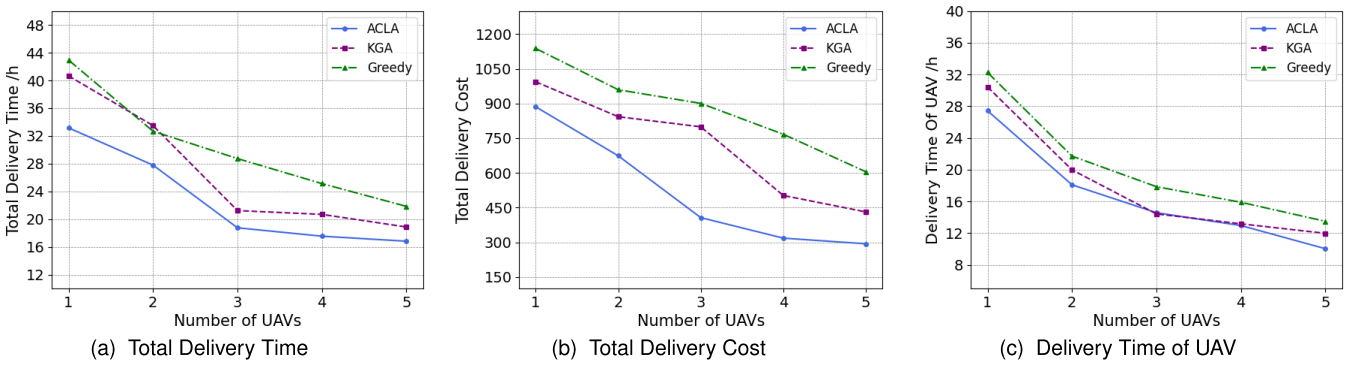


Fig. 11. The impact of different UAV numbers.

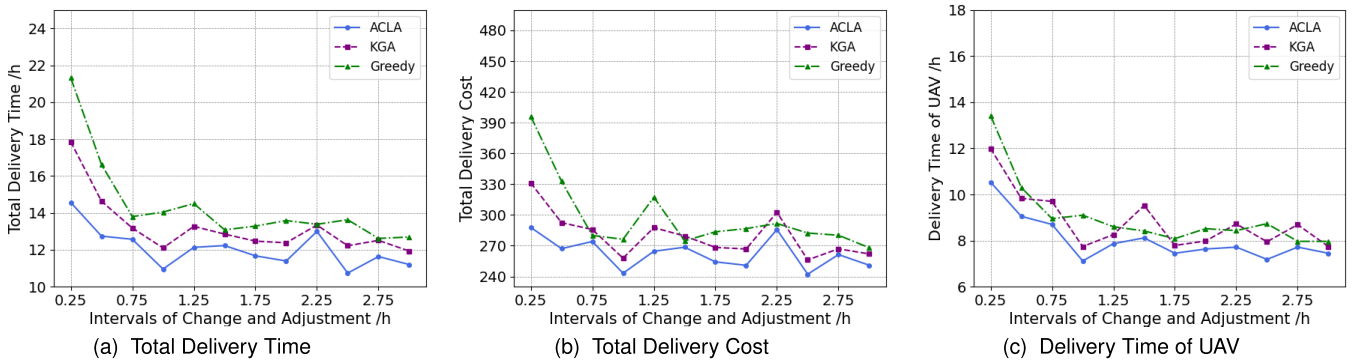


Fig. 12. The impact of different dynamic changes and adjustment cycles.

at their default values. Under tested conditions (300 customers, 40 km region, 3 UAVs), Fig. 12 demonstrates ACLA's consistent superiority over KGA and the greedy algorithm across all metrics. In Fig. 12(a), ACLA shows narrow variance in total delivery time, reflecting superior adaptability to dynamic conditions, while KGA exhibits instability and the greedy algorithm performs poorest. For UAV operation time Fig. 12(b), ACLA achieves substantial reduction through effective coordination, whereas both benchmarks display inefficient routing under shifting winds. Regarding overall cost Fig. 12(c), ACLA maintains the lowest values with minimal fluctuation, attributable to energy-efficient UAV utilization and truck collaboration, outperforming the inconsistent cost control of KGA and the non-adaptive nature of the greedy approach.

2) The Impact of Different UAV Battery Capacity: Fig. 13 examines battery capacity impact (12.5-30 kWh) with 300 customers in a 40 km area using 3 UAVs. ACLA maintains stable delivery time across capacities through adaptive task allocation Fig. 13(a), while KGA suffers from static clustering and the greedy algorithm shows inefficiency in long-range flights. In Fig. 13(b), ACLA ensures stable UAV operation via real-time energy management, contrasting with the greedy algorithm's fluctuating performance. For cost Fig. 13(c), ACLA achieves the lowest and most stable values through dual-layer optimization, outperforming KGA's high cross-region costs and the greedy method's energy waste. The results demonstrate ACLA's superior adaptability through integrated sensing, coordination, and predictive control.

TABLE III
DIFFERENT DEGREES OF CUSTOMER DYNAMISM

DI(Dynamic Intensity)	Total Delivery Time /h	Cost	CPU Time
10%	4.73	222.98	21.36
20%	4.85	226.35	22.47
30%	4.92	228.36	24.84
40%	5.32	234.96	27.55
50%	5.68	237.99	31.88
Average	5.10	230.13	25.62
Standard Deviation	0.35	5.54	3.78

3) The Impact of Different Weight Combinations: In this paper, the objective function is $minZ = \lambda_1 E_{total} + \lambda_2 T_{total}$, where λ_1 and λ_2 are the weights of the total energy consumption of the UAV E_{total} and the total delivery time T_{total} respectively, and $0 \leq \lambda_1 \leq 1$, $0 \leq \lambda_2 \leq 1$, $\lambda_1 + \lambda_2 = 1$. By adjusting the values of λ_1 and λ_2, a sensitivity analysis of the performance of the algorithm under different weights is carried out, and the results are shown in Fig. 14. A high λ_1 prioritizes UAV energy conservation at the expense of longer delivery time and higher cost. To leverage favorable winds, the algorithm may route trucks on longer paths, increasing truck travel distance and time, thereby elevating overall cost due to the high unit transport cost of trucks. When λ_1 and λ_2 are comparable, the system balances UAV energy use, delivery time, and cost by leveraging tailwinds while maintaining routing efficiency. In contrast, a small λ_1 prioritizes delivery

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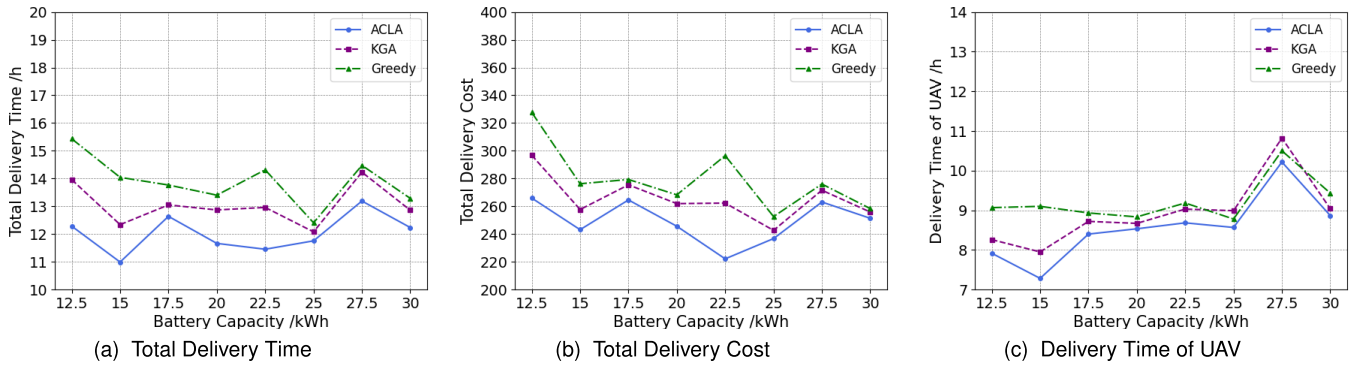


Fig. 13. The impact of different UAV battery capacity.

TABLE IV
THE DELIVERY SEQUENCE OF TRUCKS AND UAVS

Truck Parking Areas	UAV	Delivery sequence(x-coordinate, y-coordinate, cargo weight, wind speed, wind direction)
(13024,6209)	1	(13024,6209)->(12727,147,3,4.56,347.80)->(15052,1886,6,3.60,207.63)->(13024,6209)
	2	(13024,6209)->(9410,10781,5,7.85,116.98)->(13024,6209)
	3	(13024,6209)->(14780,7252,5,9.80,26.49)->(13024,6209)
		(13024,6209)->(18936,2920,11,3.40,118.79)->(13024,6209)
(7990,5238)	1	(7990,5238)->(3242,830,2,2.49,35.99)->(7990,5238)
	2	(7990,5238)->(943,6577,13,8.86,240.39)->(7990,5238)
	3	(7990,5238)->(5327,5592,3,3.23,49.27)->(7072,1606,11,4.59,317.83)->(7990,5238)
(4994,15923)	1	(4994,15923)->(5973,22448,12,5.77,339.94)->(4994,15923)
	2	(4994,15923)->(3434,18880,9,9.74,140.63)->(3506,19138,3,2.76,216.42)->(9739,18178,2,2.18,107.11)->(4994,15923)
	3	(4994,15923)->(9946,15255,15,9.58,318.10)->(4994,15923)
		(4994,15923)->(3455,14854,14,4.33,150.84)->(4994,15923)
(16661,19778)	1	(16661,19778)->(23522,18522,7,5.59,244.80)->(22402,17538,1,3.92,0.50)->(16661,19778)
		(16661,19778)->(18910,20502,2,8.54,193.41)->(15171,19611,12,6.56,227.78)->(16661,19778)
	2	(16661,19778)->(14354,22506,5,8.45,242.01)->(12228,18368,3,7.79,143.92)->(16661,19778)
		(16661,19778)->(11651,22442,8,9.84,275.26)->(10422,21995,13,7.40,199.07)->(16661,19778)
	3	(16661,19778)->(14323,16805,3,9.63,36.95)->(16775,16894,11,2.06,264.15)->(16661,19778)
		(16661,19778)->(23738,19598,5,6.26,251.26)->(16661,19778)
(20255,9113)	1	(20255,9113)->(20269,13738,3,2.83,199.93)->(22551,13681,15,7.48,278.15)->(20255,9113)
	2	(20255,9113)->(20357,10451,10,6.96,12.28)->(23212,6273,4,5.31,329.60)->(20255,9113)

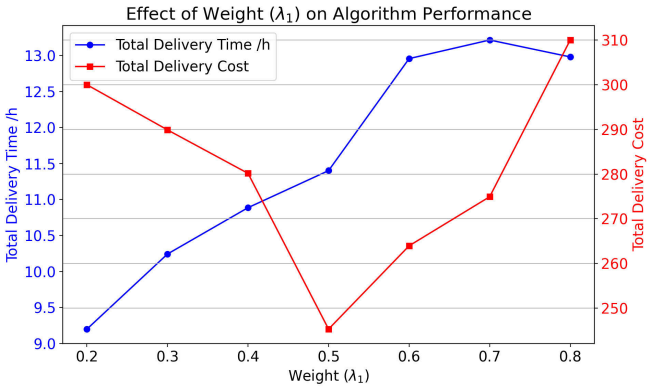


Fig. 14. The impact of different weight combinations.

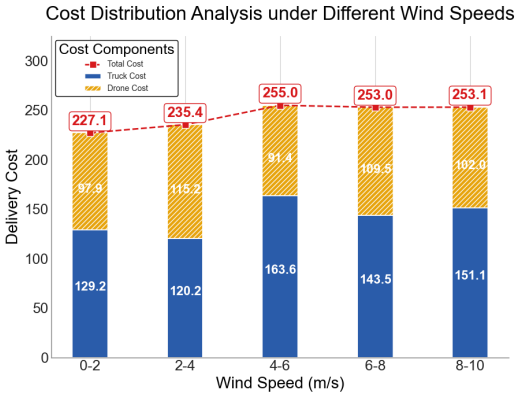


Fig. 15. Cost variation diagram of multi-wind speed gradient UAVs and truck.

speed, yielding faster service but higher energy consumption and cost as UAVs fly more direct routes against potential headwinds.

4) The Impact of Different Dynamic Intensity: This experiment employed a 20 × 20 km area with 200 customers and other parameters at default. Dynamic demand ratios

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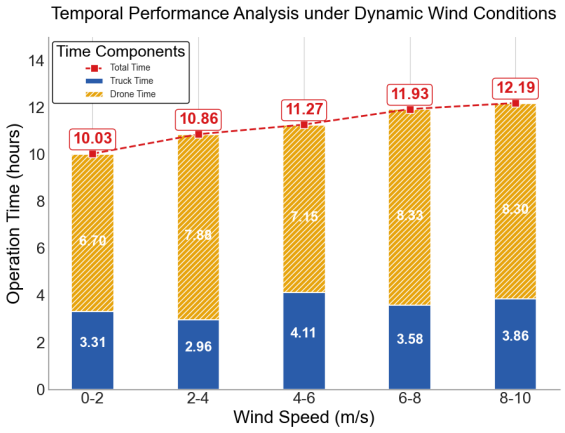


Fig. 16. Time variation diagram of multi-wind speed gradient UAVs and truck.

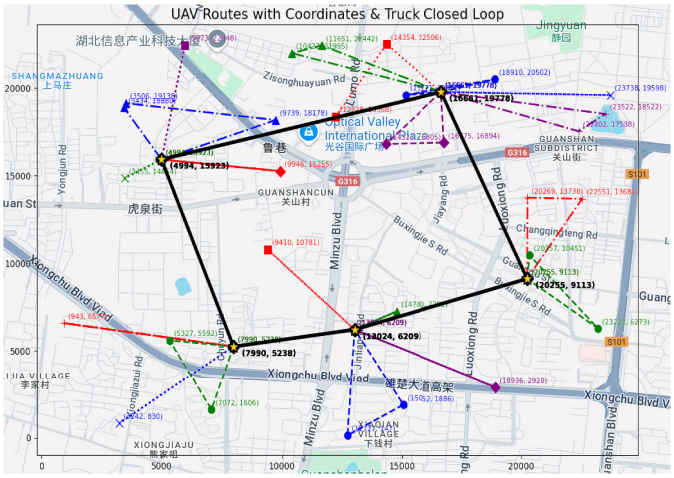


Fig. 17. Empirical case analysis on Guanggu square district (Wuhan, Hubei).

from 10% to 50% were tested, with each case independently executed 10 times and results averaged for statistical reliability (Table III). As shown in the table, rising proportions of dynamic demand increase both total delivery time and cost, due to the need for continuous real-time adjustments of truck and UAV routes, which raises operational complexity. The ACLA, however, consistently maintains stable performance across all metrics—delivery time, cost, and runtime—demonstrating robust adaptability to fluctuating demands and confirming its practical reliability.

5) Multi-Wind Speed Gradient Experiments: To evaluate ACLA's environmental adaptability, multi-level wind speed experiments (0-2 to 8-10 m/s) were conducted with 300 customers in a 40 × 40 km area using 3 UAVs. Results (Figs. 15-16) demonstrate robust performance in dynamic wind environments: total cost ranges between 227.07 and 254.98, with truck cost proportion adaptively adjusting from 56.3% and 64.1%. When wind speed increases from 0-2 to 8-10 m/s, total operation time rises by only 21.5%. Notably, under 4-6 m/s winds, UAV operation time decreases by 9.3% through intelligent scheduling, validating the algorithm's effective balance of efficiency and risk.

D. Verification Experiments in Real Scenarios

A real-world geospatial framework was established using Wuhan's Guanggu Square as the test zone, with a 25 × 25 km service area containing 30 uniformly distributed customers. Operational data were retrieved via Google Maps API, and demand was synthetically generated using demographic indices. The system, deploying 1 truck and 3 UAVs from a central depot, demonstrated marked efficiency in complex scenarios, with coordinated routes detailed in Fig. 17 and Table IV.

VI. Conclusion

This paper introduces a truck-UAV cooperative delivery model that tackles fluctuating wind conditions and dynamic customer demands to minimize energy use and total delivery time. Wind speed and direction are modeled via a Normal-Weibull-von Mises mixture, while customer demand is captured using a Poisson-density clustering approach. We propose the ACLA algorithm, which combines a Hybrid Genetic Algorithm for static planning with tabu search for real-time dynamic adjustment. Simulations show ACLA outperforms greedy and KGA methods across varying task numbers, area sizes, and UAV fleet scales, demonstrating clear adaptability and efficiency. The framework offers practical last-mile solutions for e-commerce and fresh-food logistics, fostering low-carbon, intelligent delivery systems. Future work will incorporate time-varying road networks, multimodal transport interactions, and deep reinforcement learning for enhanced robustness.

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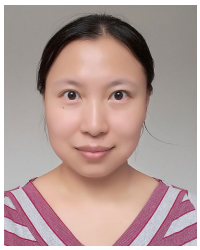
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